

U.S. Data Center Power Demand Surges Past 180 TWh — Projected to Double or Triple by 2028

Posted by u/philiejimster • • 0 points • 0 comments

Data center energy use keeps climbing fast. In 2023, U.S. data centers consumed roughly 176 TWh ($\approx 4.4\%$ of national electricity). By 2025, that's already around 183 TWh, and projections suggest 325 – 580 TWh by 2028 — about $6.7 - 12\%$ of total U.S. demand.

Load growth has tripled over the past decade and could double or triple again by 2028. Utilities in major hubs (especially Northern Virginia) are warning that new data-center demand already exceeds grid capacity by nearly 50% , driving major power-planning shifts nationwide.

Meanwhile, AI workloads and GPU-dense racks are pushing site designs toward hundreds of kW per rack, with next-gen AI campuses targeting gigawatt-scale IT loads.

I compiled several updated charts and a U.S. data-center map (from NREL / LBNL / Visual Capitalist / Pew Research 2025).

It will be interesting to see how operators, engineers, and utilities are preparing for this next wave of power demand.

Sources:

\-Pew Research Center (2025) – “What We Know About Energy Use at U.S. Data Centers Amid the AI Boom”

\-Visual Capitalist (2025) – “Charted: The Energy Demand of U.S. Data Centers (2023-2030P)”

\-Lawrence Berkeley National Laboratory (LBNL, 2024) – U.S. Data Center Energy Usage Report

Graphs are also linked here: <https://imgur.com/a/data-center-graphs-power-consumption-oBAskY5>

Comments

u/EnspiTech • 3 points • 2025-11-13 12:53 UTC

This is great. Anyone know how Texas is approaching this being that ERCOT marches to a different beat?